UNIT IV

Chapter 9

Locomotion and Movement

Chapter Outline

9.1 Types of movement
9.2 Types of muscles
9.3 Skeletal muscle
9.4 Structure of contractile proteins
9.5 Mechanism of muscle contraction
9.6 Types of skeletal muscle contraction
9.7 Skeletal system and its functions
9.8 The Axial skeleton
9.9 The Appendicular skeleton
9.10 Types of joints
9.11 Disorders of muscular and skeletal system
9.12 Benefits of regular Exercise



Leaping movement is effected by the coordination of skeletal and neuromuscular systems.

Learning Objectives:

- *Relates the structure of skeletal muscle with its function.*
- *Learns to identify bones of the skeletal system.*
- *Gains knowledge about the disorders related to muscular and skeletal systems.*
- *Understands the benefits of regular exercise.*

Have you ever wondered how a dancer performs intricate dance steps or how a swimmer skillfully does a butterfly stroke? The muscles of our body work simultaneously with one another and with the skeletal system to perform the various movements. Our muscles have two functions: to generate motion and force. All these activities are controlled and coordinated by the skeletal, muscular and nervous system. The human body is capable of a wide range of movements from the gentle blinking of eye to running a 20 km marathon. Movement of organism from one place to another in search of food, shelter, mate and to escape from predators is called locomotion. Locomotion has evolutionary significance.

9.1 Types of movement

The different types of movements that occur in the cells of our body are amoeboid, ciliary, flagellar and muscular movement.

Amoeboid movement - Cells such as macrophages exhibit amoeboid movement

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